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Summary of Theses

PART I: CURRENT MASTER'S THESIS

Title: Projected Subgradient Optimization with Polyak Steps and Langevin Perturbations for Constrained Single-Cell Representation Learning under Non-Smooth Potentials

Objective and Formulation

The project formulates the learning of low-dimensional representations from high-dimensional single-cell RNA sequencing data as a constrained, non-smooth optimization problem. The architecture is a linear autoencoder with tied, non-negative weights. The objective function $\mathcal{J}(W)$ integrates the reconstruction loss $\mathcal{L}(A, \hat{A})$ (which can be computed using either the squared Frobenius norm or the generalized Kullback-Leibler divergence), an ℓ_1 sparsity penalty, a squared Frobenius norm penalty ($\|W\|_F^2$), and optionally a Laplacian graph-based regularization:

$$\min \mathcal{J}(W) = \mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \mathcal{L}_{\text{graph}} + \lambda_1 \|W\|_1 + \lambda_2 \|W\|_F^2 \quad \text{s.t.} \quad A, \hat{A} \geq 0, W, Z \geq 0$$

Which is then decomposed into a smooth component, $h(W)$, and a non-smooth component, $c(W)$. The generalized subgradient at epoch k is then:

$$G^{(k)} = \underbrace{\nabla_W \mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \nabla_W \mathcal{L}_{\text{graph}} + 2\lambda_2 W^{(k)}}_{\nabla h(W^{(k)})} + \underbrace{\lambda_1 \text{sgn}(W^{(k)})}_{\in \partial c(W^{(k)})}$$

where sgn is the signum function applied element-wise to each weight $W_{ij}^{(k)}$.

Deterministic Optimization Dynamics

Due to the non-differentiability of the ℓ_1 penalty at the origin and the hard non-negativity constraints, the study evaluates projected subgradient descent strategies. The tests show that the scaled Polyak step-size rule achieves the best convergence by dynamically adapting to the optimality gap:

$$\alpha_k = \alpha \frac{\mathcal{J}(W^{(k)}) - \mathcal{J}^*}{c_{po} \|G^{(k)}\|_F^2}$$

Where α, c_{po} scaling factors. This rule provides stable descent trajectories and minimizes the objective without requiring manual learning-rate decay compared to diminishing, decaying and constant step sizes.

Stochastic Exploration via Langevin Dynamics

Langevin dynamics are introduced to explore the posterior parameter space via the Unadjusted Langevin Algorithm, combined with Euclidean projections onto the feasible set. The parameter update incorporates isotropic Gaussian noise $\eta^{(k)}$:

$$W^{(k+1)} = \max(0, W^{(k)} - \alpha_k G^{(k)} + \sqrt{2\alpha_k} \eta^{(k)})$$

To approximate the Bayesian Model Average without excessive memory overhead, an online ergodic average of the network weights is maintained. The averaged weights W_{Lang} are updated incrementally at each sampled step s according to:

$$W_{\text{Lang}}^{(s)} = \frac{s-1}{s} W_{\text{Lang}}^{(s-1)} + \frac{1}{s} W^{(k_s)}$$

The Langevin perturbations conflicts with non-negative projections, preventing exact sparsity and inflating parameter norms. While this limits stochastic refinement in linear models, it still marginally improves clustering on higher-dimensional datasets. The scaled Polyak step-size rule yields the main performance advantages when compared with standard PCA, NMF, and Adam-based optimization.

Title: Real-time 3D Multi-modal Multi-object Tracking via Machine Learning and Analytic Collision Risk Calculation for Autonomous Vehicles Navigation

State Estimation and Feature Fusion

The system performs 3D multi-object tracking utilizing data from one LiDAR and six cameras. State estimation was formulated via a 3D Kalman Filter using a constant linear and angular velocity dynamic model. The process noise covariance and observation noise covariance matrices were computed directly from the variance of sequential measurements. The state vector consists of:

$$s_t = (x, y, z, a, l, w, h, dx, dy, dz, da)^T$$

where x, y, z denote the target's position in the reference frame, a its orientation, l, w, h the dimensions of the bounding box, and dx, dy, dz, da the change in these parameters between two consecutive detections.

The geometric features (from LiDAR, $F_{3D} \in \mathbb{R}^{N \times 512 \times 3 \times 3}$) and appearance features (from the corresponding camera out of the 6, $F_{2D} \in \mathbb{R}^{N \times (1024+6)}$) are fused into a joint representation $F_{det}^{fused} \in \mathbb{R}^{N \times 512 \times 3 \times 3}$ using a learned multi-layer perceptron $G_1(\cdot)$. This unit serves a twofold purpose: it transforms the dimensions of the appearance features to match the geometric features, and it aligns their vectors directionally so that their pairwise addition becomes meaningful:

$$F_{det}^{fused} = G_1(F_{2D}) + F_{3D}$$

These fused features for incoming detections and existing tracks (F_{trk}^{fused}) are then processed by a secondary neural network $G_2(\cdot)$ to compute a deep affinity matrix D_{feat} , expressing the pairwise similarities for data association:

$$D_{feat} = G_2(F_{det}^{fused}, F_{trk}^{fused})$$

To address the sparsity of the true association matrix during training, the two networks are optimized using a focal loss formulation, which down-weights the overwhelming number of negative (non-matching) associations:

$$FL(p_t) = -\alpha \cdot (1 - p_t)^\gamma \cdot \log(p_t)$$

where p_t represents the predicted probability of the correct class, α is the ratio between negative and positive associations, and γ is a focusing parameter. Empirical results show that this deep affinity matrix is more representative when compared to simpler standard techniques such as cosine similarity.

Analytic Collision Risk Calculation

The Collision State Probability was calculated by modeling the spatial overlap between the autonomous vehicle and tracked obstacles. The Minkowski sum of the object bounding geometries was computed to produce a collision octagon. The coordinate system is transformed based on the obstacle's position and orientation, so that the principal axes of the obstacle's Gaussian covariance align with the orthogonal axes of the reference frame. The collision risk was evaluated analytically by integrating over the independent Gaussian state distributions of the objects within this polygonal domain. By decomposing the boundary into regions of positive and negative contribution, the total probability at time T_p was computed using the Gaussian error function (erf):

$$P(C, T_p) = \sum_{i=0}^{i \leq m} P(C_i, T_p) - \sum_{i=m}^{i \leq n} P(C_i, T_p)$$

where $P(C_i, T_p)$ represents the probability contribution of an individual line segment C_i of the collision octagon. By decomposing the octagon into these line segments, the contribution is calculated by numerically integrating the Gaussian error function over the boundaries of the segment. Line segments are assigned a positive or negative contribution depending on whether they lie below or above the horizontal axis, while segments nearly parallel to the vertical axis are excluded. If the octagon lies below the x -axis, the positive and negative contributions are reversed to correctly capture the enclosed probability volume:

$$P(C_i, T_p) = \frac{1}{\sqrt{8\pi}\sigma_x} \int_{x_{l,i}}^{x_{u,i}} \text{erf}\left(\frac{m_i x + b_i}{\sqrt{2}\sigma_y}\right) e^{-\frac{x^2}{2\sigma_x^2}} dx$$